

settings

tab_name
wdi_2016
wdi_csv_open_csv
wdi_csv_parquet
wdi_csv_text
wdi_gs_debug
wdi_opencsv_gs
wdi_opencsv_text
wdi_opencsv_text_partitions

Took 0 sec. Last updated by anonymous at September 02 2025, 4:09:45 AM.

%hive
DESCRIBE wdi_csv_text;

FINISHED

settings

col_name	data_type	comment
year	int	
countryname	string	
countrycode	string	
indicatorname	string	
indicatorcode	string	
indicatorvalue	float	

Took 0 sec. Last updated by anonymous at September 02 2025, 4:09:59 AM. (outdated)

%hive
SELECT * FROM wdi_csv_text LIMIT 10;

FINISHED

settings

year	countryname	countrycode	indicatorname	indicatorcode	indicatorvalue
1960	Aruba	ABW	Listed domestic companies	total	null
1960	Aruba	ABW	Binding coverage	all products (%)	null
1960	Aruba	ABW	GDP (current US\$)	NY.GDP.MKTP.CD	0.0
1960	Aruba	ABW	Poverty headcount ratio at national poverty lines (% of population)	SI.POV.NAHC	0.0
1960	Aruba	ABW	PNG	bonds (AMT	null
indicatorvalue		float			

Took 0 sec. Last updated by anonymous at September 02 2025, 4:09:59 AM. (outdated)

%hive
SELECT * FROM wdi_csv_text LIMIT 10;

FINISHED

settings

year	countryname	countrycode	indicatorname	indicatorcode	indicatorvalue
1960	Aruba	ABW	Listed domestic companies	total	null
1960	Aruba	ABW	Binding coverage	all products (%)	null
1960	Aruba	ABW	GDP (current US\$)	NY.GDP.MKTP.CD	0.0
1960	Aruba	ABW	Poverty headcount ratio at national poverty lines (% of population)	SI.POV.NAHC	0.0
1960	Aruba	ABW	PNG	bonds (AMT	null
1960,Barbados,BRB,People practicing open defecation, urban (% of urban population),SH.STA.ODFC.UR.ZS,0.0					

Took 21 sec. Last updated by anonymous at September 02 2025, 9:34:28 PM. (outdated)

Took 0 sec. Last updated by anonymous at September 02 2025, 10:11:27 PM.

FINISHED

%hive
CREATE EXTERNAL TABLE wdi_opencsv_gs (
year STRING,
country_name STRING,
country_code STRING,
indicator_name STRING,
indicator_code STRING,
value STRING

READY

```
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.OpenCSVSerde'
WITH SERDEPROPERTIES (
  "separatorChar" = ",",
  "quoteChar" = "\"",
  "escapeChar" = "\\\"
)
STORED AS TEXTFILE
LOCATION 'gs://jarvis_dataeng_bootcamp/wdi_csv_text';
```

```
2.1G .
2.1G total

#clear fs cache
echo 3 | sudo tee /proc/sys/vm/drop_caches

#bash row count
aasadityashuklw@jarvis-hadoop-cluster-m:~$ cd ~/wdi_csv_text
aasadityashuklw@jarvis-hadoop-cluster-m:~/wdi_csv_text$ date +%s && cat * | wc && date +%s
1756242785
21759408 198019710 2171529559
1756242794

21759408 → number of lines (rows) across all files
198019710 → number of words (not very relevant for CSV, but wc counts space-separated tokens)
2171529559 → number of bytes (total size of all files combined)

The timestamps before and after (1756242785 → 1756242794) show the operation took ~9 seconds to process all 11 CSV chunks.
```

Took 0 sec. Last updated by anonymous at September 02 2025, 9:21:59 PM.

FINISHED

```
-- LIST DISTINCT INDICATOR CODE, ORDERED ALPHABETICALLY, LIMITED TO 20 RESULTS
SELECT distinct(indicatorcode)
FROM wdi_csv_text
ORDER BY indicatorcode;
```

HIVE JOB FINISHED

settings

tab_name

wdi_2016
wdi_csv_open_csv
wdi_csv_parquet
wdi_csv_text
wdi_gs_debug
wdi_opencsv_gs
wdi_opencsv_text
wdi_opencsv_text_partitions

Took 0 sec. Last updated by anonymous at September 02 2025, 4:09:45 AM.

```
%hive
DESCRIBE wdi_csv_text;
```

FINISHED

1960,Aruba,ABW,Market capitalization of listed domestic companies (current US\$),CM.MKT.LCAP.CD,0.0

Took 0 sec. Last updated by anonymous at September 02 2025, 9:32:32 PM.

```
%hive
SELECT *
FROM wdi_gs_debug
WHERE line LIKE '%(% of urban population)%';
```

HIVE JOB FINISHED

settings

line

1960,Argentina,ARG,Access to non-solid fuel, urban (% of urban population),EG.NSF.ACCS.UR.ZS,0.0
1960,American Samoa,ASM,Access to electricity, urban (% of urban population),EG.ELC.ACCS.UR.ZS,0.0
1960,Belgium,BEL,People practicing open defecation, urban (% of urban population),SH.STA.ODFC.UR.ZS,0.0
1960,Bangladesh,BGD,Access to non-solid fuel, urban (% of urban population),EG.NSF.ACCS.UR.ZS,0.0
1960,Bangladesh,BGD,Population in the largest city (% of urban population),EN.URB.LCTY.UR.ZS,20.521181
1960,Bosnia and Herzegovina,BIH,Population in the largest city (% of urban population),EN.URB.LCTY.UR.ZS,27.664469
1960,Burkina Faso,BUR,Access to non-solid fuel, urban (% of urban population),EG.NSF.ACCS.UR.ZS,0.0

```
2.1G .
2.1G total

#clear fs cache
echo 3 | sudo tee /proc/sys/vm/drop_caches

#bash row count
aasadityashuklw@jarvis-hadoop-cluster-m:~$ cd ~/wdi_csv_text
aasadityashuklw@jarvis-hadoop-cluster-m:~/wdi_csv_text$ date +%s && cat * | wc && date +%s
1756242785
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The timestamps before and after (1756242785 → 1756242794) show the operation took ~9 seconds to process all 11 CSV chunks.
```

Took 0 sec. Last updated by anonymous at September 02 2025, 9:21:59 PM.

FINISHED

-- LIST DISTINCT INDICATOR CODE, ORDERED ALPHABETICALLY, LIMITED TO 20 RESULTS

HIVE JOB FINISHED

```
SELECT distinct(indicatorcode)
FROM wdi_csv_text
```

Took 21 sec. Last updated by anonymous at September 02 2025, 8:38:45 PM. (outdated)

```
SELECT count(countryname) FROM wdi_csv_text;
```

HIVE JOB FINISHED

settings

_c0
21759408

```
CREATE EXTERNAL TABLE wdi_gs_debug (
  line STRING
)
STORED AS TEXTFILE
LOCATION '/user/aasadityashuklw/wdi_csv_text/';
```

FINISHED

Took 0 sec. Last updated by anonymous at September 02 2025, 9:31:44 PM.

```
SELECT * FROM wdi_gs_debug LIMIT 10;
```

FINISHED

settings

line
1960,Aruba,ABW,Listed domestic companies, total,CM.MKT.LDOM.NO,0.0
1960,Aruba,ABW,Binding coverage, all products (%),TM.TAX.MRCH.BC.ZS,0.0
1960,Aruba,ABW,GDP (current US\$),NY.GDP.MKTP.CD,0.0
1960,Aruba,ABW,Poverty headcount ratio at national poverty lines (% of population),SI.POV.NAHC,0.0
1960,Aruba,ABW,PNG, bonds (AMT, current US\$),DT.AMT.PNGB.CD,0.0
1960,Aruba,ABW,Technicians in R&D (per million people),SP.POP.TECH.RD.P6,0.0
1960,Aruba,ABW,Number of surgical procedures (per 100 000 population),SH.SGR.PROC.P5,0.0

```
country_name    STRING,
country_code    STRING,
indicator_name   STRING,
indicator_code   STRING,
value           STRING
)
PARTITIONED BY (year STRING)
STORED AS TEXTFILE
```

Took 1 sec. Last updated by anonymous at September 02 2025, 11:41:01 PM.

```
SET hive.exec.dynamic.partition = true;
SET hive.exec.dynamic.partition.mode = nonstrict;
INSERT OVERWRITE TABLE wdi_opencsv_text_partitions
PARTITION (year)
SELECT
country_name,
country_code,
indicator_name,
indicator_code,
value,
year
FROM wdi_opencsv_text;
```

HIVE JOB FINISHED

Took 1 min 13 sec. Last updated by anonymous at September 02 2025, 11:42:18 PM.

Took 0 sec. Last updated by anonymous at September 02 2025, 11:07:57 PM.

```
--Viewing the metadata and storage information from the table 'wdi_opencsv_gs'
DESCRIBE FORMATTED wdi_opencsv_gs;
```

FINISHED

settings

col_name	data_type	comment
# col_name	data_type	comment
year	string	from deserializer
country_name	string	from deserializer
country_code	string	from deserializer
indicator_name	string	from deserializer
indicator_code	string	from deserializer
value	string	from deserializer
	null	null

1.0782688	2015	Canada
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Took 19 sec. Last updated by anonymous at September 02 2025, 11:34:16 PM. (outdated)

Took 0 sec. Last updated by anonymous at September 02 2025, 11:38:08 PM.

FINISHED

Took 0 sec. Last updated by anonymous at September 02 2025, 11:39:34 PM.

FINISHED

DROP TABLE IF EXISTS wdi_opencsv_text_partitions;

FINISHED

settings

indicator_code

%)
15+
Austria (current US\$)
Atlas method (current US\$)
Belgium (current US\$)
Canada (current US\$)
Australia (current US\$)
(% of urban population)

Took 20 sec. Last updated by anonymous at September 02 2025, 10:49:32 PM.

```
%hive
--Compare execution time between 'wdi_opencsv_text' and 'wdi_csv_text'
SELECT COUNT(country name) FROM wdi_opencsv_text
```

HIVE JOB FINISHED

```
-- Crating a new view to simplify querying 'wdi_opencsv_gs'
DROP VIEW IF EXISTS wdi_opencsv_text_view;
CREATE VIEW wdi_opencsv_text_view AS
SELECT *
FROM wdi_opencsv_text;
```

FINISHED

Took 0 sec. Last updated by anonymous at September 02 2025, 11:15:49 PM.

Took 0 sec. Last updated by anonymous at September 02 2025, 11:18:59 PM.

FINISHED

```
SELECT DISTINCT indicator_name, indicator_code
FROM wdi_opencsv_text
WHERE indicator_name LIKE "%GDP growth (annual %)%";
```

HIVE JOB FINISHED

settings

indicator_name	indicator_code
GDP growth (annual %)	NY.GDP.MKTP.KD.ZG

settings

_c0
43518816

Took 33 sec. Last updated by anonymous at September 02 2025, 10:49:58 PM.

```
SELECT COUNT(countryName) FROM wdi_csv_text
```

HIVE JOB FINISHED

1.0782688	2015	Canada
-----------	------	--------

Took 19 sec. Last updated by anonymous at September 02 2025, 11:34:16 PM. (outdated)

Took 0 sec. Last updated by anonymous at September 02 2025, 11:38:08 PM.

FINISHED

Took 0 sec. Last updated by anonymous at September 02 2025, 11:39:34 PM.

FINISHED

DROP TABLE IF EXISTS wdi_opencsv_text_partitions;

FINISHED

```
ROW_NUMBER() OVER (
  PARTITION BY countryname
  ORDER BY indicatorvalue DESC
) AS rn
FROM wdi_csv_parquet
WHERE indicatorcode = 'NY.GDP.MKTP.KD.ZG'
) t
WHERE rn = 1;
```

VERTICES: 00/02 [=====] 50% ELAPSED TIME: 10.06 s

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	5	5	0	0	0	0
Reducer 2		container	INITED	1	0	0	1	0	0

VERTICES: 01/02 [=====] 83% ELAPSED TIME: 11.82 s

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	5	5	0	0	0	0
Reducer 2		container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 02/02 [=====] 100% ELAPSED TIME: 12.14 s

settings

1.07826882015Canada

Took 19 sec. Last updated by anonymous at September 02 2025, 11:34:16 PM. (outdated)

Took 0 sec. Last updated by anonymous at September 02 2025, 11:38:08 PM.

FINISHED

Took 0 sec. Last updated by anonymous at September 02 2025, 11:39:34 PM.

FINISHED

DROP TABLE IF EXISTS wdi_opencsv_text_partitions;

FINISHED

SELECT COUNT(countryName) FROM wdi_csv_parquet;

FINISHED

settings

_c0
43518816

```
country_name STRING,
country_code STRING,
indicator_name STRING,
indicator_code STRING,
value STRING
)
PARTITIONED BY (year STRING)
```

STORED AS TEXTFILE

Took 1 sec. Last updated by anonymous at September 02 2025, 11:41:01 PM.

```
SET hive.exec.dynamic.partition = true;
SET hive.exec.dynamic.partition.mode = nonstrict;
INSERT OVERWRITE TABLE wdi_opencsv_text_partitions
PARTITION (year)
SELECT
country_name,
country_code,
indicator_name,
indicator_code,
value,
year
FROM wdi_opencsv_text;
```

HIVE JOB FINISHED

Took 1 min 13 sec. Last updated by anonymous at September 02 2025, 11:42:18 PM.

gdp_growth_value	year	country_name
1.0782688	2015	Canada
1.0782688	2015	Canada

Took 19 sec. Last updated by anonymous at September 02 2025, 11:34:16 PM. (outdated)

Took 0 sec. Last updated by anonymous at September 02 2025, 11:38:08 PM.

FINISHED

Took 0 sec. Last updated by anonymous at September 02 2025, 11:39:34 PM.

FINISHED

```
DROP TABLE IF EXISTS wdi_opencsv_text_partitions;
```

FINISHED

Took 0 sec. Last updated by anonymous at September 03 2025, 12:20:55 AM. (outdated)

Took 0 sec. Last updated by anonymous at September 03 2025, 12:31:01 AM.

FINISHED

Took 0 sec. Last updated by anonymous at September 03 2025, 12:40:35 AM.

FINISHED

settings

HIVE JOB FINISHED

gdp_growth_value	year	country_name
1.0782688	2015	Canada
1.0782688	2015	Canada

Took 19 sec. Last updated by anonymous at September 03 2025, 12:33:56 AM. (outdated)

```
SELECT indicatorvalue AS GDP_growth_value,
year,
countryname
FROM wdi_csv_parquet
WHERE countryname = 'Canada'
AND year = '2015'
AND indicatorcode = 'NY.GDP.MKTP.KD.ZG';
```

HIVE JOB FINISHED

settings

gdp_growth_value	year	countryname
1.0782688	2015	Canada
1.0782688	2015	Canada

Took 19 sec. Last updated by anonymous at September 03 2025, 12:33:56 AM. (outdated)

```
SELECT indicatorvalue AS GDP_growth_value,
year,
```

HIVE JOB FINISHED

```
countryname
FROM wdi_csv_parquet
WHERE countryname = 'Canada'
AND year = '2015'
AND indicatorcode = 'NY.GDP.MKTP.KD.ZG';
```

settings

gdp_growth_value	year	countryname
1.0782688	2015	Canada
1.0782688	2015	Canada

Took 0 sec. Last updated by anonymous at September 03 2025, 12:20:55 AM. (outdated)

Took 0 sec. Last updated by anonymous at September 03 2025, 12:31:01 AM.

FINISHED

Took 0 sec. Last updated by anonymous at September 03 2025, 12:40:35 AM.

FINISHED

settings

HIVE JOB FINISHED

gdp_growth_value	year	country_name
1.0782688	2015	Canada
1.0782688	2015	Canada

Took 19 sec. Last updated by anonymous at September 03 2025, 12:33:56 AM. (outdated)

```
SELECT indicatorvalue AS GDP_growth_value,
       year,
       countryname
FROM wdi_csv_parquet
WHERE countryname = 'Canada'
AND year = '2015'
AND indicatorcode = 'NY.GDP.MKTP.KD.ZG';
```

HIVE JOB FINISHED

settings

gdp_growth_value	year	countryname
1.0782688	2015	Canada
1.0782688	2015	Canada

Took 0 sec. Last updated by anonymous at September 02 2025, 11:15:49 PM.

Took 0 sec. Last updated by anonymous at September 02 2025, 11:18:59 PM.

FINISHED

```
SELECT DISTINCT indicator_name, indicator_code
FROM wdi_opensv_text
WHERE indicator_name LIKE "%GDP growth (annual %)%";
```

HIVE JOB FINISHED

settings

indicator_name	indicator_code
GDP growth (annual %)	NY.GDP.MKTP.KD.ZG

Took 19 sec. Last updated by anonymous at September 03 2025, 12:33:56 AM. (outdated)

```
SELECT indicatorvalue AS GDP_growth_value,
       year,
       countryname
FROM wdi_csv_parquet
WHERE countryname = 'Canada'
AND year = '2015'
AND indicatorcode = 'NY.GDP.MKTP.KD.ZG';
```

HIVE JOB FINISHED

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	5	5	0	0	0	0
Reducer 2	container	SUCCEEDED	1	1	0	0	0	0
VERTICES: 02/02 [=====>>] 100% ELAPSED TIME: 12.14 s								

settings

Took 7 sec. Last updated by anonymous at September 03 2025, 1:06:48 AM.

- Hive on CSV/Text table: 18–22 seconds
- Hive on Parquet table: 11–14 seconds
- SparkSQL on Parquet table: 7 seconds

Took 0 sec. Last updated by anonymous at September 03 2025, 1:08:58 AM.

Sort GDP by country and year

Took 0 sec. Last updated by anonymous at September 03 2025, 1:09:44 AM.

```
-- Sort GDP by country and year
SELECT
  country_name,
  year,
  indicator_code,
  value AS GDP_growth_value
FROM wdi_opencsv_text
WHERE indicator_code = 'NY.GDP.MKTP.KD.ZG'
```

```
year,
indicatorvalue,
ROW_NUMBER() OVER (
  PARTITION BY countryname
  ORDER BY indicatorvalue DESC
) AS rn
FROM wdi_csv_parquet
WHERE indicatorcode = 'NY.GDP.MKTP.KD.ZG'
) t
WHERE rn = 1;
```

settings

country_name	year	GDP_growth_value
Afghanistan	2009	21.020649
Albania	1999	13.501173
Algeria	1963	34.31373
American Samoa	1965	0.0
Andorra	2003	12.02392
Angola	2007	22.593054
Antigua and Barbuda	2006	13.3764
Arab World	1976	12.896334

Took 0 sec. Last updated by anonymous at September 03 2025, 12:20:55 AM. (outdated)

Took 0 sec. Last updated by anonymous at September 03 2025, 12:31:01 AM.

Took 0 sec. Last updated by anonymous at September 03 2025, 12:40:35 AM.

settings

gdp_growth_value	year	country_name
1.0782688	2015	Canada

Took 19 sec. Last updated by anonymous at September 03 2025, 12:33:56 AM. (outdated)

```
SELECT indicatorvalue AS GDP_growth_value,
  year,
  countryname
FROM wdi_csv_parquet
WHERE countryname = 'Canada'
AND year = '2015'
AND indicatorcode = 'NY.GDP.MKTP.KD.ZG';
```

settings

gdp_growth_value	year	countryname
1.0782688	2015	Canada
1.0782688	2015	Canada